Power BOQ — Project P-379

Proposed B+G+8+R Commercial & Residential Building
Plot 6731315, Al Barsha South Third, Dubai · Authority: DEWA

Job No · FA_P379
Drawing Set · P-001 … P-300 (14 sheets)
Date · 16.01.2026
Revision · v1.0 (reconciled)

RECONCILIATION NOTE — corrections applied vs source files

- **ESMDB-G feeder cable**: source PDF section 4 listed `1×4C 70 mm²` FR; section 8.7 listed `1×4C 300 mm²` FR. Resolved to **1×4C 300 mm² FR + 1×1C 150 mm² ECC** — 70 mm² is undersized by ~3× for ~338 A load. `[XLSX wins / PDF self-conflict]`
- **LVP-01 → SMDB-1F…8F riser cable**: source XLSX sized at `4C × 300 mm²`; source PDF sized at `1×4C 150 mm² + 1C 70 mm²` ECC. Resolved to **1×4C 150 mm² XLPE/SWA/PVC + 1×1C 70 mm² ECC** — 150.86 kW @ 400 V ≈ 256 A; 150 mm² is the textbook size. `[PDF wins — XLSX oversized]`
- **Fire-pump feeder cable**: source PDF `1×4C 185 mm²` FR; source XLSX `2×4C 300 mm²` FR. Resolved to **1×4C 185 mm² FR + 1×1C 95 mm² ECC** — 167 A starting; 185 mm² FR adequate. `[PDF wins — XLSX oversized]`
- **Fire-pump breaker**: source PDF `350 A TP`; source XLSX `300 A TP MCCB (ATS)`. Resolved to **350 A TP MCCB (with shunt-trip)** — typical for fire pump starter. `[PDF wins]`
- **Drawing date**: source XLSX self-conflict (Project Info 16.01.2026 vs Drawing Register 12.02.2026). Resolved to **16.01.2026** across the document. `[PDF wins]`

1. Project Summary

Item	Value
Project	Proposed B+G+8+R Commercial & Residential Building
Plot No.	6731315
Location	Al Barsha South Third, Dubai, UAE
Owner	Qutaiba Ameen Abdal Kija
Architect	Engr. Samer Mahmoud Ajami (Reg. 105181)
Structural Engineer	Engr. Mohamad Maher Myaser Jabban (Reg. 101079)
MEP Consultant	Future Art Engineering Consultancy
Job No.	FA_P379 / CRs B/010/25
Drawing Set	P-001 … P-300 (14 sheets, Power Layout)
Drawing Date	16.01.2026
Authority	DEWA (Dubai Electricity & Water Authority)
Floors	14 levels: UG (basement) → G → 1F–8F → R → UR

2. Incoming Supply & Transformers

Item	Description	Qty	Unit
2.1	DEWA HV incoming service connection (RMU side, by DEWA)	1	lot
2.2	HV/LV distribution transformer to suit Building Maximum Demand of **~1,597 kW** (DF 0.80, TCL 2,117.06 kW). Sizing per DEWA approval. SOURCE	1	lot
2.3	HV cable from RMU to transformer LV side, complete with terminations and earthing	1	lot
2.4	Transformer room civil/MEP coordination — ventilation, drainage, fire-rated enclosure	1	lot

Transformer kVA, oil/dry type, vector group and impedance to be finalised per DEWA load letter response. Sizing target: ≥ 1,597 kW continuous + 25% spare ≈ 2,000 kVA dry-type.

3. LV Panels (Main Distribution)

Item	Panel	TCL (kW)	MD (kW)	Incomer / Bus	Qty
3.1	**LVP-01** — Main LV Panel #1 (essential + non-essential)	1,206.89	965.51	4P ACB, ≥ 2,000 A; bus 2,500 A	1
3.2	**LVP-02** — Main LV Panel #2 (services + tenant feeders)	910.17	631.34	4P ACB, ≥ 1,250 A; bus 1,600 A	1
3.3	Outgoing MCCB feeder for SMDB-1F (150.86 kW · ~256 A)	4P MCCB, 400 A frame, 250 A trip			8
3.4	Outgoing MCCB feeder for SMDB-G (60.55 kW)	4P MCCB, 160 A frame, 125 A trip			1
3.5	Outgoing MCCB feeder for SMDB-RF (248.63 kW)	4P MCCB, 630 A frame, 500 A trip			1
3.6	Outgoing MCCB feeder for SMDB-EV (88.00 kW)	4P MCCB, 250 A frame, 160 A trip			1
3.7	Outgoing MCCB feeder for ESMDB-G (199.13 kW)	4P MCCB, 630 A frame, 400 A trip CORRECTED			1
3.8	Outgoing MCCB feeder for ESMDB-RF (82.69 kW)	4P MCCB, 250 A frame, 160 A trip			1
3.9	Outgoing MCCB feeder for Fire Pump (98 kW · 167 A start)	4P MCCB, **350 A TP** with shunt-trip CORRECTED			1

4. Sub-Main Distribution Boards (SMDB)

Item	SMDB Tag	Location	Connected Load (kW)	Incomer	Qty

4.1	SMDB-1F	1st Floor (electrical riser)	150.86	4P MCCB 250 A	1
4.2	SMDB-2F	2nd Floor	150.86	4P MCCB 250 A	1
4.3	SMDB-3F	3rd Floor	150.86	4P MCCB 250 A	1
4.4	SMDB-4F	4th Floor	150.86	4P MCCB 250 A	1
4.5	SMDB-5F	5th Floor	150.86	4P MCCB 250 A	1
4.6	SMDB-6F	6th Floor	150.86	4P MCCB 250 A	1
4.7	SMDB-7F	7th Floor	150.86	4P MCCB 250 A	1
4.8	SMDB-8F	8th Floor	150.86	4P MCCB 250 A	1
4.9	SMDB-G	Ground Floor (lobby / retail)	60.55	4P MCCB 125 A	1
4.10	SMDB-RF	Roof Floor (chillers / lift machine room)	248.63	4P MCCB 500 A	1
4.11	SMDB-EV	Basement (EV charging)	88.00	4P MCCB 160 A	1
4.12	ESMDB-G	Ground (essential — life safety + fire)	199.13	4P MCCB 400 A	1
4.13	ESMDB-RF	Roof (essential — fire pump panel feeder)	82.69	4P MCCB 160 A	1
Total SMDB connected load			**1,837.59**		

5. Distribution Boards (DB)

DB-level itemisation drawn from XLSX sheet "SMDB to DB" (108 rows · total cable run 1,578 m). Per-DB summary below; full DB list in Section 8 cable schedule.

Item	Fed From	DBs (typical use)	Cable run (m)	Qty
5.1	SMDB-1F … SMDB-8F	DB-Apt-A, DB-Apt-B, DB-Corridor (per typical floor)	~120 / floor	24
5.2	SMDB-G	DB-Lobby, DB-Retail-1, DB-Retail-2	~85	3
5.3	SMDB-RF	DB-Chiller, DB-Lift-MR, DB-Pump, DB-AHU-Roof	~140	4
5.4	SMDB-EV	DB-EV-1, DB-EV-2 (basement charging)	~95	2

5.5	ESMDB-G	DB-FA, DB-Emergency-Lighting, DB-Smoke-Vent	~110	3
5.6	ESMDB-RF	DB-Fire-Pump-Aux, DB-Stair-Pressurisation	~70	2
Total DB count / total SMDB→DB cable			**~1,578**	**~38**

6. Mechanical & Service Equipment Loads

Connection (power supply & control wiring) for mechanical / service equipment. Equipment supply by others. Largest single load: FAHU at 174.66 kW. SOURCE 15 line items per source PDF section 6 — full schedule from MEP coordination.

Item	Equipment	Connected Load (kW)	Feed	Qty
6.1	FAHU (Fresh Air Handling Unit)	174.66	SMDB-RF	1
6.2	Chillers / chilled-water pumps (per HVAC drawings)	see HVAC	SMDB-RF	lot
6.3	Domestic water pumps (booster set)	see Plumbing	SMDB-G	lot
6.4	Sewage / sump pumps	see Plumbing	ESMDB-G	lot
6.5	Fire pump (jockey + main + diesel cooling)	98.00	ESMDB-RF (Fire Pump panel)	1
6.6	Stair pressurisation fans	see HVAC	ESMDB-G	lot
6.7	Smoke-extract fans (basement & corridors)	see HVAC	ESMDB-G	lot
6.8	Lift drives (passenger + service)	see Lift sub-cont.	SMDB-RF	lot
6.9	Kitchen exhaust / make-up air fans (retail)	see HVAC	SMDB-G	lot
6.10	EV charger feeders (7 kW & 22 kW points)	88.00	SMDB-EV	lot
6.11–6.15	Misc. service equipment (gate motor, water tank level, BMS field, irrigation pump, garbage chute fan)	balance	various	lot
Standby generator allowance (essential loads, gen-fed)		**23.00**	**ESMDB feeders**	

7. Power Outlets & Accessories

19 line items per source PDF section 7 — apartment and common-area accessories. Final quantities per architectural finishes schedule. SOURCE

Item	Description	Unit	Indicative Qty
7.1	Single 13 A switched socket outlet (BS 1363, white finish)	no	~340
7.2	Twin 13 A switched socket outlet	no	~280

7.3	13 A SSO with USB-A/USB-C (apartment living/bedrooms)	no	~110
7.4	20 A DP switched outlet for AC indoor unit	no	~95
7.5	20 A DP switched outlet for water heater	no	~50
7.6	32 A cooker / oven outlet with isolator	no	~25
7.7	Weatherproof IP55 13 A SSO (balconies, podium, roof)	no	~60
7.8	Floor-mounted service box (lobby / retail counters)	no	~18
7.9	Connection unit for ceiling fan / extract fan (3 A FCU)	no	~75
7.10	Shaver socket outlet (bathrooms)	no	~40
7.11	Doorbell push + chime kit (apartments)	set	25
7.12	1-gang / 2-gang lighting switches (white grid)	no	~520
7.13	Dimmer switch (1-gang, 250 W)	no	~60
7.14	2-way / intermediate switch (corridors, stairs)	no	~45
7.15	Occupancy / PIR sensor (corridors, BOH)	no	~30
7.16	Isolator (4P 32 A) for outdoor mech. equipment	no	~22
7.17	EV charger socket outlet — 7 kW Mode 3 Type 2 (basement)	no	8
7.18	EV charger socket outlet — 22 kW Mode 3 Type 2 (basement)	no	2
7.19	Final-fix testing & commissioning of accessories	lot	1

8. Cables — Main Distribution Cable Schedule

LV → SMDB cable schedule, drawn from XLSX sheet "LV to SMDB" (26 itemised rows, total **1,056 m**) and reconciled against PDF sections 8.1–8.10. All runs include +10% installation allowance.

Item	From	To	Load (kW)	Current (A)	Cable Size (Phase + ECC)	Length incl. +10% (m)
8.1 LVP-01 → typical-floor SMDB risers (1F–8F)						
8.1.1	LVP-01	SMDB-1F	150.86	256	1×4C 150 mm² XLPE/SWA/PVC + 1×1C 70 mm² ECC CORRECTED	22
8.1.2	LVP-01	SMDB-2F	150.86	256	1×4C 150 mm² XLPE/SWA/PVC + 1×1C 70 mm² ECC	26
8.1.3	LVP-01	SMDB-3F	150.86	256	1×4C 150 mm² XLPE/SWA/PVC + 1×1C 70 mm² ECC	29
8.1.4	LVP-01	SMDB-4F	150.86	256	1×4C 150 mm² XLPE/SWA/PVC + 1×1C 70 mm² ECC	33

8.1.5	LVP-01	SMDB-5F	150.86	256	1×4C 150 mm² XLPE/SWA/PVC + 1×1C 70 mm² ECC	36
8.1.6	LVP-01	SMDB-6F	150.86	256	1×4C 150 mm² XLPE/SWA/PVC + 1×1C 70 mm² ECC	40
8.1.7	LVP-01	SMDB-7F	150.86	256	1×4C 150 mm² XLPE/SWA/PVC + 1×1C 70 mm² ECC	43
8.1.8	LVP-01	SMDB-8F	150.86	256	1×4C 150 mm² XLPE/SWA/PVC + 1×1C 70 mm² ECC	47
Subtotal — 8 typical-floor risers (LVP-01 → SMDB-1F…8F)						**276**
8.2 LVP-02 → roof / ground / EV SMDBs						
8.2.1	LVP-02	SMDB-RF	248.63	422	2×4C 150 mm² XLPE/SWA/PVC + 1×1C 150 mm² ECC	62
8.2.2	LVP-02	SMDB-G	60.55	103	1×4C 50 mm² XLPE/SWA/PVC + 1×1C 25 mm² ECC	14
8.2.3	LVP-02	SMDB-EV	88.00	149	1×4C 70 mm² XLPE/SWA/PVC + 1×1C 35 mm² ECC	28
Subtotal — services / roof / EV feeders						**104**
8.3 Essential feeders (gen-backed) — Fire Rated						
8.3.1	LVP-02	ESMDB-G	199.13	338	1×4C 300 mm² FR + 1×1C 150 mm² ECC CORRECTED	18
8.3.2	LVP-02	ESMDB-RF	82.69	140	1×4C 70 mm² FR + 1×1C 35 mm² ECC	58
8.3.3	ESMDB-RF	Fire Pump panel	98.00	167 / 350 start	1×4C 185 mm² FR + 1×1C 95 mm² ECC CORRECTED	36
Subtotal — essential feeders						**112**
8.4 Auxiliary & balance feeders (per XLSX itemisation)						
8.4.x	LVP / SMDB	Auxiliary loads (HVAC isolators, lift main switch, BMS panel, generator control, lighting risers, etc.)		balance	per cable schedule (XLSX rows 17–26)	~564
TOTAL — LV → SMDB cable (with +10% allowance)						**~1,056**

SMDB → DB cable schedule (108 itemised runs in XLSX sheet "SMDB to DB")	Total length incl. +10% (m)
TOTAL — SMDB → DB cable	**1,578**

9. Containment (Cable Tray, Trunking, Conduit)

12 line items per source PDF section 9. Sized to take cable schedule above + 30% spare. Hot-dip galvanised finish (basement/roof), pre-galvanised (apartment / dry areas). SOURCE

Item	Description	Unit	Indicative Qty
9.1	600 mm wide hot-dip galvanised cable tray (riser shaft & basement)	m	~180
9.2	450 mm wide HDG cable tray (corridor mains)	m	~220
9.3	300 mm wide HDG cable tray (branch routes)	m	~340
9.4	200 mm wide HDG cable tray	m	~260
9.5	100 mm wide HDG cable tray	m	~180
9.6	Cable tray accessories — bends, tees, reducers, copplers (≈ 15% of tray length)	lot	1
9.7	Cable tray supports — threaded rod, channel, anchors	lot	1
9.8	50×50 mm GI trunking (DB sub-mains)	m	~140
9.9	100×100 mm GI trunking (sub-main risers)	m	~85
9.10	HG steel conduit Ø 25/32/40 mm with accessories (concealed final circuits)	m	~3,400
9.11	HG steel conduit Ø 20 mm (accessory drops)	m	~2,800
9.12	Fire-rated cable cleats & spacers (FR feeders, sec 8.3)	lot	1

10. Earthing & Lightning Protection

7 line items per source PDF section 10. System: TN-S, building earth ≤ 1 Ω. SOURCE

Item	Description	Unit	Qty
10.1	Main earth bar (250 × 50 × 6 mm copper) in LV room	no	2
10.2	Earth electrode pit with 2.4 m × Ø 19 mm copper-bonded rod, inspection chamber	no	8
10.3	70 mm² bare copper earth conductor (electrode interconnection / risers)	m	~240
10.4	25 mm² PVC-insulated earth conductor (DB sub-circuits)	m	~520
10.5	Lightning protection air terminals (ESE / Franklin, per consultant spec)	no	6
10.6	Down-conductor 50 mm² bare copper, full building height	m	~240

10.7	Earthing testing & certification (DEWA acceptance)	lot	1

11. Metering & Monitoring

6 line items per source PDF section 11. **132 kWh meters** (apartments + common services + retail + landlord) — per DEWA tenant-metering standard. SOURCE

Item	Description	Unit	Qty
11.1	Single-phase DEWA-approved kWh meter, apartment tenants (CT-operated)	no	~120
11.2	Three-phase DEWA-approved kWh meter, retail / common services / landlord	no	~12
11.3	Tenant meter cabinet — modular, lockable, ventilated	no	8
11.4	Multi-function digital meter (LV panel incomers / outgoers, MODBUS to BMS)	no	12
11.5	Current transformers — 250/5, 400/5, 630/5, 2000/5 (per panel schedule)	set	1
11.6	Wiring, terminations & commissioning of metering system, DEWA acceptance	lot	1
Total kWh meters (apartments + common + landlord)			**132**

12. Summary of Electrical Loads

Panel	Total Connected Load — TCL (kW)	Demand Factor	Maximum Demand — MD (kW)
LVP-01 (residential risers + EV)	1,206.89	0.80	965.51
LVP-02 (services + retail + roof + essential)	910.17	0.69	631.34
Building Total	**2,117.06**	**0.755**	**~1,597**
of which: standby generator allowance (essential, gen-fed)	23.00	included in ESMDB feeders	

Distribution Cable Totals	Length (m, incl. +10%)	Source
LV → SMDB main feeders	~1,056	XLSX sheet "LV to SMDB" (26 rows)
SMDB → DB sub-mains	~1,578	XLSX sheet "SMDB to DB" (108 rows)
Total main + sub-main cable	**~2,634**	**both sheets**

Reconciliation source: P-379_POWER_BOQ.pdf (deliverable BOQ, 12 sections) and "Extracted Data for Electrical BOQ and Estimation.xlsx" (working take-off, 6 sheets). Verdict-driven: PDF wins on building totals +

typical-cable-size accuracy + section coverage; XLSX wins on per-circuit cable lengths + drawing register + floor schedule.

Items pending source itemisation (marked SOURCE in document): Section 6 mechanical equipment list, Section 7 accessory final-fix counts, Section 9 containment lengths, Section 10 earthing items, Section 11 metering breakdown. These quantities are indicative — to be confirmed from architectural finishes schedule, MEP coordination drawings and tenant-meter layout before pricing finalisation.

Cross-check applied: ESMDB-G feeder size now consistent in Sections 3 (400 A MCCB), 4 (incomer note) and 8.3.1 (cable size) — the section 4 vs 8.7 mismatch in the source PDF is resolved.

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